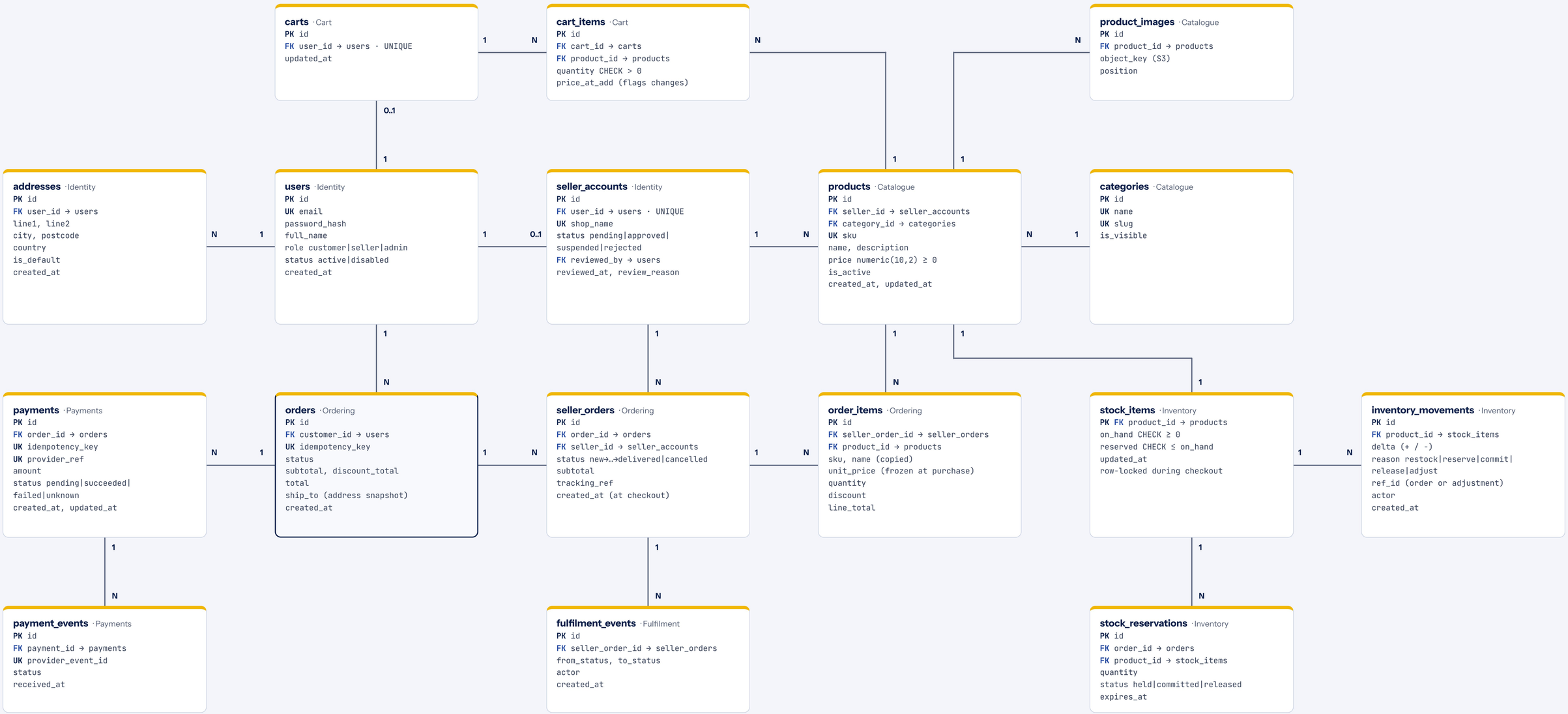
Mini Marketplace — entity relationship diagram (PostgreSQL)

20 tables in one shared PostgreSQL schema, grouped by bounded context (shown after each table name). Foreign keys may cross contexts for integrity, but only the owning context's code writes to its tables.

PK primary key · **FK** foreign key · **UK** unique · line labels show cardinality: **1** one, **N** many, **0..1** optional



PLATFORM TABLES · NO FOREIGN KEYS: THEY POINT AT ANY ROW BY TYPE + ID

outbox_events · Platform (outbox)
PK id
aggregate_type, aggregate_id
event_type
payload jsonb
created_at
published_at (NULL = unsent)

processed_messages · Platform (worker)
PK (message_id, handler)
processed_at
Each handler records its own copy, so one handler finishing never makes another skip the message.

audit_log · Audit
PK id
actor_type user|system
actor_id
action
entity_type, entity_id
before jsonb
after jsonb
created_at

Integrity rules the schema enforces

- UNIQUE orders.idempotency_key — pressing checkout twice returns the same order
- UNIQUE payments.idempotency_key — a retried payment never charges twice
- stock_items CHECK reserved ≤ on_hand, row locked (SELECT … FOR UPDATE) during checkout — no overselling
- order_items copy sku, name and unit_price — later price changes never alter past orders
- stock_reservations.expires_at — a scheduled job releases holds from abandoned checkouts
- processed_messages (message_id, handler) — a redelivered message is skipped · outbox_events saved in the same transaction